# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

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| **Course Code:** | CSA10 / CSA1063 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO2 Statement:** | *Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills* | | |
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### Code of Conduct

*I Tamil Elakiya s (192524443) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Tamil Elakiya S

# Digital farmer Marketplace for Agricultural Product Trading

## 1. Study and Analyse the Scenario

### 1.1 Introduction

The Digital Farmer Marketplace is a software-based platform designed to improve the agricultural trading process by connecting farmers directly with buyers. The system replaces the traditional marketing approach with an online marketplace where farmers can list their products, buyers can browse and purchase products, and administrators can monitor the overall operation. The proposed solution aims to improve transparency, increase farmers' income, and simplify agricultural product trading through digital technology.

### 1.2 Problem Statement

Agriculture plays a vital role in the country's economy, but many farmers still depend on traditional methods of selling their produce. They often rely on middlemen, wholesalers, and local markets to reach customers. Due to the involvement of intermediaries, farmers receive lower profits while buyers pay higher prices. In addition, the existing system lacks transparency, efficient communication, and proper record management.

The proposed Digital Farmer Marketplace addresses these issues by providing an online platform where farmers and buyers can interact directly. Farmers can upload product details, manage inventory, and receive online payments, while buyers can search products, compare prices, place orders, and track deliveries. This digital approach improves efficiency and ensures a fair trading environment for all stakeholders.

### 1.3 Business Domain

The project belongs to the **Agriculture and E-Commerce** domain. It combines agricultural marketing with information technology to provide an online marketplace for buying and selling agricultural products. The platform also integrates digital payment services and logistics support to improve the overall agricultural supply chain.

### 1.4 Stakeholders

The major stakeholders involved in the Digital Farmer Marketplace are:

- **Farmers:** Register on the platform, upload products, manage inventory, and receive payments.
- **Buyers:** Search products, place orders, make payments, and provide feedback.
- **Retailers and Wholesalers:** Purchase agricultural products in bulk for resale.
- **Delivery Partners:** Transport products from farmers to buyers.
- **Platform Administrator:** Manage users, products, transactions, and system operations.
- **Banks and Payment Gateway Providers:** Process secure online payments.
- **Government and Agricultural Departments:** Promote digital agriculture and monitor fair trading practices.

### 1.5 Existing Challenges

The current agricultural trading system faces several challenges that affect both farmers and buyers.

- Farmers depend heavily on middlemen, reducing their profits.
- Buyers have limited access to fresh products directly from farmers.
- Product pricing lacks transparency.
- Manual record-keeping increases the possibility of errors.
- Payments are often delayed.
- Farmers have limited access to wider markets.
- Communication between farmers and buyers is inefficient.
- Perishable products may be wasted due to slow sales and poor inventory management.

### 1.6 Objectives

The Digital Farmer Marketplace is designed with the following objectives:

- Develop a secure online marketplace for agricultural products.
- Connect farmers directly with buyers.
- Eliminate unnecessary intermediaries.
- Ensure transparent pricing.
- Enable secure online payment processing.

- Simplify product listing, ordering, and inventory management.
- Maintain digital transaction records.
- Improve market accessibility for farmers.
- Reduce food wastage through efficient order management.
- Build a scalable and user-friendly software system.

# 2. Identify Functional and Non-Functional Requirements

## 2.1 Introduction

Requirements define the expected functionalities and quality attributes of a software system. They help developers understand what the system should do and how well it should perform. The requirements of the Digital Farmer Marketplace are classified into **Functional Requirements** and **Non-Functional Requirements**.

Functional requirements describe the services provided by the system, whereas non-functional requirements specify the performance, security, usability, and other quality characteristics that ensure the system operates efficiently.

## 2.2 Functional Requirements

Functional requirements specify the core features that the Digital Farmer Marketplace must provide to support agricultural product trading.

- **FR1: User Registration :** The system shall allow farmers, buyers, and administrators to register by providing the required personal information.
- **FR2: User Login :** The system shall authenticate registered users using a valid username and password before granting access.
- **FR3: Profile Management :** Users shall be able to update their personal information, contact details, and account settings.

- **FR4: Product Management :** Farmers shall be able to add, edit, update, and delete agricultural product listings.
- **FR5: Product Search :** Buyers shall be able to search products using keywords, categories, or product names.
- **FR6: Product Filtering :** The system shall allow buyers to filter products based on price, category, location, and availability.
- **FR7: Shopping Cart :** The system shall allow buyers to add products to a shopping cart before placing an order.
- **FR8: Order Placement :** Buyers shall be able to place orders for selected agricultural products.
- **FR9: Payment Processing :** The system shall support secure online payment methods for completing transactions.
- **FR10: Order Tracking :** Users shall be able to monitor the status of placed orders until delivery.
- **FR11: Notifications :** The system shall notify users about order confirmations, payment status, and delivery updates.
- **FR12: Review and Rating :** Buyers shall be able to provide ratings and feedback for purchased products.
- **FR13: Report Generation :** The administrator shall generate reports related to users, products, orders, and transactions.
- **FR14: User Management :** The administrator shall verify, approve, or deactivate user accounts whenever necessary.
- **FR15: Complaint Management :** Users shall be able to submit complaints, and administrators shall resolve them through the platform.

## 2.3 Non-Functional Requirements

Non-functional requirements define the quality standards that the Digital Farmer Marketplace should achieve.

**Performance**

The system should process user requests quickly and support multiple users simultaneously without significant delays.

**Security**

User credentials, payment information, and personal data should be protected through secure authentication and encryption mechanisms.

**Reliability**

The system should operate consistently with minimal failures and provide accurate transaction processing.

**Usability**

The user interface should be simple, intuitive, and easy to understand, allowing users with basic digital knowledge to operate the platform.

**Scalability**

The application should support future expansion by accommodating more users, products, and transactions without affecting performance.

**Maintainability**

The software should have a modular structure, making it easier to update, debug, and introduce new features.

**Availability**

The system should remain accessible to users at all times, except during scheduled maintenance.

# 3. **Identify Actors and Use Cases**

## 3.1 Introduction

Actors and use cases help define how different users interact with the proposed system. An **actor** represents a person or external system that communicates with the software, while a **use case** describes a specific function or service provided by the system. Identifying actors and their associated use cases helps developers understand the system's functional requirements and design the software accordingly.

## 3.2 Primary and Secondary Actors

The Digital Farmer Marketplace consists of both **primary** and **secondary** actors.

**Primary Actors :**

Primary actors directly interact with the system to perform various operations.

- Farmer
- Buyer
- Administrator

**Secondary Actors :**

Secondary actors support the operation of the system but do not directly control its core functionalities.

- Payment Gateway
- Delivery Partner
- Government/Agricultural Department

## 3.3 List the Major Use Cases Associated with Each Actor

**Farmer**

The farmer is the main supplier of agricultural products on the platform.

**Major Use Cases:**

- Register an account
- Login to the system
- Manage profile
- Add new products
- Update product details
- Delete unavailable products
- Manage inventory
- View customer orders
- Receive payment notifications
- Track product deliveries
- View sales history
- Logout

**Buyer**

The buyer purchases agricultural products directly from farmers.

**Major Use Cases:**

- Register an account
- Login to the system
- Search products
- Filter products
- View product details
- Add products to cart
- Place orders
- Make online payment
- Track orders
- View order history

**Administrator**

The administrator manages and monitors the entire marketplace.

**Major Use Cases:**

- Login
- Verify users
- Manage farmer accounts
- Manage buyer accounts
- Monitor product listings
- Monitor transactions
- Resolve complaints
- Generate reports
- Manage system settings
- Logout

**Payment Gateway**

The payment gateway supports secure online financial transactions.

**Major Use Cases:**

- Process payments
- Verify payment status
- Generate payment confirmation
- Maintain transaction records

**Delivery Partner**

The delivery partner transports agricultural products to buyers.

**Major Use Cases:**

- Receive delivery requests
- Update delivery status
- Confirm successful delivery

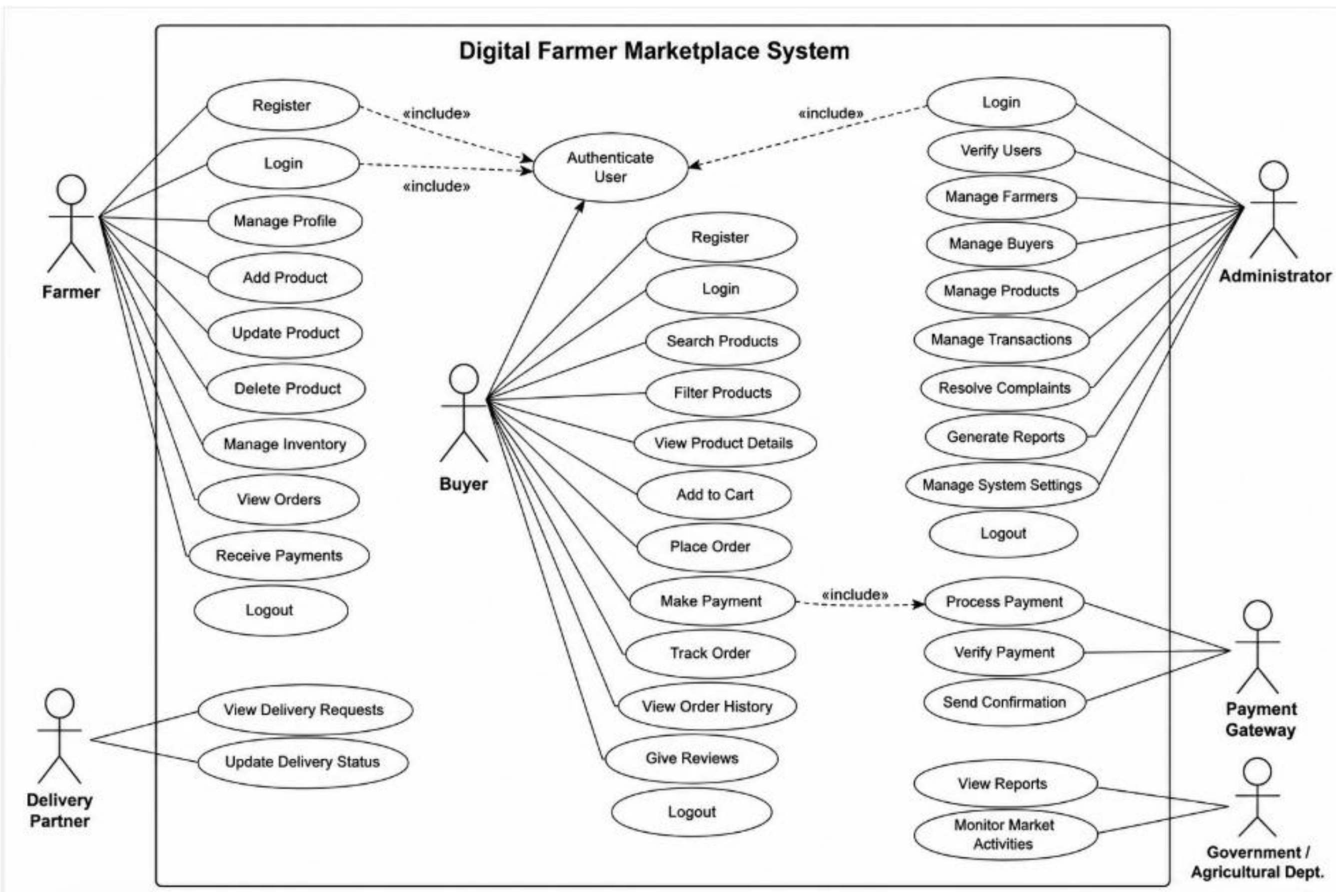
**Government / Agricultural Department**

Government agencies monitor agricultural activities and support digital farming initiatives.

**Major Use Cases:**

- View agricultural reports
- Monitor market activities
- Analyze trading statistics

## 3.4 Use Case Diagram :



# 4. Prepare the Software Requirement Specification (SRS)

## 4.1 Introduction

### 4.1.1 Purpose of the Document

The purpose of this Software Requirement Specification (SRS) document is to define the complete requirements of the **Digital Farmer Marketplace for Agricultural Product Trading**. This document describes the functional and non-functional requirements, system features, user interactions, data requirements, and external interfaces required for developing the proposed system.

The SRS acts as a communication document between stakeholders, developers, testers, and project managers. It ensures that the developed software meets the expectations of farmers, buyers, administrators, and other stakeholders.

### 4.1.2 Overview of the System

The Digital Farmer Marketplace is a web-based platform designed to connect farmers directly with consumers and other buyers. The system eliminates unnecessary intermediaries and provides a transparent environment for agricultural product trading.

Farmers can register on the platform, upload product details, manage inventory, and receive payments directly from buyers. Buyers can search agricultural products, compare prices, place orders, and track deliveries. The administrator manages users, transactions, and overall system operations.

The system aims to improve farmer income, provide fair pricing to consumers, reduce food wastage, and promote digital transformation in agriculture.

## 4.2 Problem Description

In the existing agricultural trading system, farmers often receive low profits despite producing quality products. The involvement of middlemen reduces transparency because farmers sell their products at lower prices while consumers purchase them at higher prices.

The lack of direct communication between farmers and buyers creates several problems such as:

- Low income for farmers.
- Higher product prices for consumers.
- Limited market access for farmers.
- Lack of transparent pricing information.
- Delayed payments.
- Difficulty in managing product availability and sales records.
- Food wastage due to unsold agricultural products.

The proposed Digital Farmer Marketplace addresses these problems by creating a direct digital connection between farmers and buyers.

### 4.3 Scope of the System

The scope of the Digital Farmer Marketplace includes the complete process of agricultural product trading.

The system covers:

**Farmer Services**

- Farmer registration and authentication.
- Product listing and inventory management.
- Order management.
- Payment tracking.
- Sales history maintenance.

**Buyer Services**

- Product searching and filtering.
- Viewing product details.
- Placing orders.
- Online payment.
- Order tracking.
- Product reviews.

**Administrative Services**

- User verification.
- Product monitoring.
- Transaction management.
- Complaint handling.
- Report generation.

**System Scope**

The platform focuses on improving agricultural marketing efficiency, reducing dependency on middlemen, and providing a reliable digital marketplace for agricultural products.

### 4.4 Data Requirements

The system requires proper storage and management of different types of data.

**User Data**

Includes:

- User ID
- Name
- Contact details
- Address
- Login credentials
- User role

**Product Data**

Includes:

- Product ID
- Product name
- Category
- Quantity
- Price
- Images

- Availability status

**Order Data**

Includes:

- Order ID
- Buyer details
- Product details
- Order status
- Delivery information

**4.5 External Interface Requirements**

**User Interface**

The system should provide:

- Simple navigation.
- Responsive design.
- Easy-to-understand forms.
- Dashboard for different users.

**Hardware Interface**

The system should support:

- Desktop computers.
- Smartphones.
- Tablets.

**Software Interface**

The system may interact with:

- Database management systems.
- Payment gateways.

- Notification services.
- Mapping services.

**Communication Interface**

The system requires internet connectivity for communication between users and the server.

## 5. Identify Assumptions and Constraints

### 5.1 Introduction

Assumptions and constraints define the conditions and limitations considered during the development of the **Digital Farmer Marketplace for Agricultural Product Trading**. Assumptions represent expected conditions for successful system operation, while constraints represent factors that may affect development and implementation.

### 5.2 Assumptions

The following assumptions are considered while developing the system:

- Users such as farmers and buyers have access to internet connectivity and digital devices.
- Farmers provide accurate information about products, prices, and availability.
- Users are willing to adopt digital methods for agricultural trading.
- Secure online payment services are available for transactions.
- Delivery services are available for transporting products.
- The database and server infrastructure are sufficient for storing and managing system data.
- Stakeholders provide feedback for future improvements.

### 5.3 Constraints

**Technical Constraints**

- The system depends on internet availability, which may be limited in rural areas.
- Data security and privacy must be maintained to protect user information.
- The system should support future growth in users and transactions.

**Operational Constraints**

- Some farmers may have limited digital knowledge and require training.
- Verification of product quality and authenticity may be challenging.
- Efficient delivery of perishable products can be difficult in remote locations.

**Economic Constraints**

- Development, maintenance, and hosting costs may affect implementation.
- Payment gateway charges may increase operational expenses.

**Legal Constraints**

- The system must comply with data privacy and agricultural trading regulations.
- Online transactions must follow financial security standards.

**Time Constraints**

- The project must be completed within the given development period.
- Advanced features may require additional time for implementation and testing.

## 6. Validate Requirement Completeness and Consistency

### 6.1 Introduction

Requirement validation ensures that all identified requirements are correct, complete, consistent, and suitable for developing the Digital Farmer Marketplace system. This process helps identify missing requirements, unclear statements, and conflicts before the software development phase begins.

### 6.2 Verification of Stakeholder Requirements

The requirements are reviewed to ensure that the needs of all stakeholders are addressed.

- **Farmers:** Product listing, inventory management, order handling, and payment tracking requirements are included.
- **Buyers:** Product search, ordering, payment, and delivery tracking requirements are covered.
- **Administrators:** User management, monitoring, and reporting requirements are included.
- **Payment and Delivery Services:** Transaction processing and delivery management requirements are considered.

### 6.3 Checking Requirement Quality

**Ambiguity Check**

Requirements are reviewed to ensure they have clear meanings and do not contain confusing statements. Each requirement is written in a simple and specific manner to avoid misunderstanding between stakeholders and developers.

**Redundancy Check**

Duplicate requirements are identified and removed to prevent repetition and maintain a clear requirement document.

**Consistency Check**

All requirements are checked to ensure they do not conflict with each other. Functional and non-functional requirements are aligned to support the overall system objectives.

**Missing Requirement Analysis**

The requirement document is reviewed to ensure important features such as:

- User authentication
- Product management
- Order processing
- Payment handling
- Data security
- Report generation.

### 6.4 Improvements to Enhance Requirement Quality

- Include continuous feedback from farmers and buyers during development.
- Add detailed security requirements for protecting user and payment data.
- Define performance requirements with measurable values.
- Include future requirements such as mobile application support, multilingual features, and AI-based recommendations.